

CNN-Adaptive

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1 Method 1: Clothoid Path

1.1 Path Generation

The reference path $P(X, Y) = 0$ transitions from a straight line to a circle of radius R . Let s denote the arc length and L the transition length.

Clothoid Spiral ($0 \leq s \leq L$): Using the standard Fresnel Formulation:

$$a = \sqrt{RL\pi},$$

the path is defined as:

$$X(t) = a \int_0^t \cos\left(\frac{\pi u^2}{2}\right) du \quad (1)$$

$$Y(t) = a \int_0^t \sin\left(\frac{\pi u^2}{2}\right) du \quad (2)$$

1.2 Optimal Point Identification

The optimal point (X^*, Y^*) is defined as the orthogonal projection of the vehicle position (X_v, Y_v) onto the path.

The optimal parameter t^* is obtained by solving:

$$t^* = \arg \min_t (X_v - X(t))^2 + (Y_v - Y(t))^2 \quad (3)$$

The first order optimality condition is:

$$\frac{d}{dt} ((X_v - X(t))^2 + (Y_v - Y(t))^2) \Big|_{t=t^*} = 0 \quad (4)$$

Which simplifies to:

$$(X_v - X(t^*))\dot{X}(t^*) + (Y_v - Y(t^*))\dot{Y}(t^*) = 0 \quad (5)$$

And using equations (1) and (2):

$$(X_v - X(t^*)) \cos\left(\frac{\pi(t^*)^2}{2}\right) + (Y_v - Y(t^*)) \sin\left(\frac{\pi(t^*)^2}{2}\right) = 0 \quad (6)$$

The solution t^* is obtained numerically (e.g., Newton-Raphson), and:

$$X^* = X(t^*), \quad Y^* = Y(t^*)$$

1.3 Error Extraction

At (X^*, Y^*) , the reference states are:

$$\psi_{\text{des}} = \frac{\pi(t^*)^2}{2} \quad (7)$$

$$(8)$$

Lateral Error:

$$D = \sqrt{(X_v - X^*)^2 + (Y_v - Y^*)^2} \quad (9)$$

$$e_y = \text{sgn}((X_v - X^*)P_Y - (Y_v - Y^*)P_X) \cdot D \quad (10)$$

Heading Error:

$$e_\psi = \text{normalize}(\psi_v - \psi_{\text{des}}) \quad (11)$$

2 Method 2: Simple Polynomial Path (Cubic Approximation)

2.1 Path Generation

For small deflection angles, the linear increase in curvature can be modeled using a cubic polynomial:

$$Y(X) = kX^3 \quad (12)$$

where k is a scaling constant. The first and second derivatives are:

$$Y'(X) = 3kX^2 \quad (13)$$

$$Y''(X) = 6kX \quad (14)$$

2.2 Curvature Analysis

The exact curvature formula is:

$$\kappa(X) = \frac{Y''(X)}{(1 + (Y'(X))^2)^{3/2}} = \frac{6kX}{(1 + 9k^2X^4)^{3/2}} \quad (15)$$

For $X \approx 0$ (the start of the transition), the denominator is approximately 1, yielding:

$$\kappa(X) \approx 6kX \quad (16)$$

This confirms that the curvature increases linearly with the coordinate X .

2.3 Optimal Point Identification

The optimal point (X^*, Y^*) for a vehicle at (X_v, Y_v) is found by minimizing:

$$X^* = \arg \min_X (X - X_v)^2 + (kX^3 - Y_v)^2 \quad (17)$$

The optimality condition (where the distance vector is orthogonal to the tangent) results in a 5th-degree polynomial:

$$(X^* - X_v) + 3k(X^*)^2(k(X^*)^3 - Y_v) = 0 \quad (18)$$

2.4 Error Extraction

The desired heading ψ_{des} at the optimal point is:

$$\psi_{\text{des}} = \text{atan}(3k(X^*)^2) \quad (19)$$

The lateral error e_y and heading error e_ψ are calculated using the same geometric logic as in Method 1.